Gradient Boosting to Address Statistical Problems Arising from Non-Linkage of Census Bureau Datasets

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Abstract

This article introduces the twangRDC package, which contains functions to address non-linkage in US Census Bureau datasets. The Census Bureau's Person Identification Validation System facilitates data linkage by assigning unique person identifiers to federal, third party, decennial census, and survey data. Not all records in these datasets can be linked to the reference file and as such not all records will be assigned an identifier. This article is a tutorial for using the twangRDC to generate nonresponse weights to account for non-linkage of person records across US Census Bureau datasets.

Keyword*:* non-response, gradient boosting, weighting, R, Census data

[*] Any opinions and conclusions expressed herein are those of the authors and do not represent the views of the U.S. Census Bureau. This paper does not use any confidential Census Bureau data.

1. Introduction

The US Census Bureau's Person Identification Validation System (PVS) facilitates data linkage by assigning unique person identifiers, known as Protected Identification Keys (PIKs) to decennial census and survey data. The PVS uses probabilistic matching to assign a unique PIK for each person based on personally identifiable information (PII) from secure reference files containing data from the Social Security Administration Numerical Identification file and other federal files. PIKs are linking keys that can be used to match person records across datasets Wagner, Lane *et al.* (2014). For example, using a PIK, a person record in the 2000 decennial census can be linked to the 2010 decennial census.

Not all records can be linked to the reference file and as such not all records will be assigned a PIK. Non-linkage may occur if the PII used to match to the reference file is of insufficient quality (e.g., missing information), has insufficiently unique identifying information (e.g., a common name, such as John Smith), or if there is no matching record in the Social Security reference file (e.g., undocumented residents). PIKs are assigned for the large majority of recent decennial census records. A decennial census record must have a PIK in order to link it to administrative records that contain other information, such as mortality or annual residence.

The **twangRDC** package contains functions to address non-linkage, which may systematically affect some records creating bias in population estimates, and is freely distributed under GPL-3 on CRAN (Cefalu and Sullivan 2021). It is a streamlined version of the **twang** R package (Ridgeway, McCaffrey, Morral, Cefalu, Burgette, Pane, and Griffin 2021) and was created specifically for use in the Federal Statistical Research Data Centers (FSRDCs). FSRDCs are restricted data enclaves run by the Census Bureau to promote research use of data products from the Census and other federal statistical agencies. This streamlining was necessary because the original **twang** package was not designed to analyze the large datasets that are available in the FSRDCs.

The **twangRDC** package utilizes gradient boosted models to estimate weights for non-linkage and comparison group construction (Natekin and Knoll 2013). Gradient boosting is an iterative machine learning technique that combines many weak prediction models to approximate the true underlying model, which is likely to be much more complex. The **twangRDC** package uses **xgboost** for gradient boosting, which offers an optimized gradient boosting library and parallel computation to make weight construction for large datasets within FSRDCs more efficient (Chen and Guestrin 2016).

Results using **twangRDC** will not necessarily be reproduced using the **twang** R package due to important differences in implementation. The **twangRDC** package allows for much larger datasets, such as those used in population sciences, and many more covariates. Users should note that with smaller datasets, the **twang** package is computationally more efficient. This article uses non-linkage within Orleans Parish, LA, as an example analysis. The provided code and results generate weights for person-records in Orleans Parish such that those with linked records are representative of their Census tract population.

2. Methodology

The **twangRDC** package uses gradient boosted models to derive weights for nonequivalent groups using the methodology of McCaffrey, Ridgeway, and Morral (2004). The algorithm alternates between adding iterations to the gradient boosted model (increasing its complexity) and evaluating balance between the two groups on the covariates. The algorithm automatically stops when additional iterations no longer improve balance. This tutorial focuses on generating weights to account for non-linkage. The package can also be used to create a comparison group for a selected set of person records. Details regarding comparison group construction are available in the package help files and vignette.

2.1. Non-Linkage of US Census Bureau Data

The **twangRDC** package generates weights to account for non-linkage such that person records with PIKs are representative of the population. It does this by using standard nonresponse weights, where nonresponse is defined by non-linkage (Särndal and Lundström 2005). This is achieved by weighting person records with PIKs by the inverse probability of having a PIK, conditional on other known information. As a note, although designed for non-linkage, the **twangRDC** can be used for general nonresponse and missing data.

The steps of the algorithm are described here assuming the user specifies strata in the model, but the steps are the same without strata by considering the data coming from a single stratum. Guidance on specific decisions relevant for each step is provided throughout the examples.

1. Calculate the population-level means of the covariates within strata. If weights are specified by the user, the population-level means are calculated accounting for the weights. There are no restrictions on the types of covariates included.
2. Initialize a gradient boosted model that includes all covariates and the strata as a factor.
3. Add `iters.per.step` iterations to the gradient boosted model.
4. Calculate the means of the covariates within strata among person records with PIK using weights based on the current model.
5. Calculate the standardized differences of the covariates between weighted person records with PIK versus the population within strata as calculated in step #1.
6. Summarize the model fit by taking the maximum of the absolute standardized differences of the covariates within strata.
7. Check if the average of the maximum absolute standardized differences across strata has been minimized. If a minimum has been achieved, stop. Otherwise, return to step #3.

3. Accounting for Non-Linkage within Orleans Parish, LA

3.1. Data

A simulated data file was created to mirror the structure of restricted versions of the 2000 decennial census available through the FSRDC network. We built the file exclusively from public use data. It contains simulated records for residents of Orleans Parish. Each simulated record includes basic individual demographic characteristics and basic household characteristics. Each record also includes two important indicators, one to simulate whether the record received a PIK and another to denote whether the individual lived in Orleans Parish.

The data was created by extracting all "short form" variables for households and individuals from the 5% Integrated Public Use Microdata Sample (IPUMS) of the 2000 decennial census for Orleans Parish and for a selection of other southern metropolitan areas (Ruggles, McCaa, Sobek, and Cleveland (2015)). We simulated assignment of households to census tracts and attached tract identifiers and characteristics. Public use data do not include PIK assignment, so we estimated a predicted probability of receiving a PIK by using estimated regression parameters from Bond, Brown, Luque, O'Hara *et al.* (2014). We added a random error to the predicted probability so that the PIK assignment status is not deterministic and converted the predicted probability into a dichotomous PIK indicator.

Lastly, we sampled the data to shrink the size of the dataset. The sampling of person records is not necessary in the FSRDCs, where we expect much larger datasets, and was only performed to ensure that the example dataset could be distributed with the **twangRDC** package. The simulated file contains person records for 8,893 residents of Orleans Parish, LA.

Table 1 provides a description of the data element included in the simulated data file. This file is available to users within the **twangRDC** package.

Data element	Description	Labels and Coding
Metarea	A categorical variable for metropolitan area	Atlanta, GA (52) Memphis, TN/AR/MS (492) New Orleans, LA (556) Washington, DC/MD/VA (884)
c00_age12	A categorical variable for age in years at the 2000 decennial census.	0 to 2 years old (1) 3 to 5 years old (2) 6 to 9 years old (3) 10 to 14 years old (4) 15 to 18 years old (5) 19 to 24 years old (6) 25 to 34 years old (7) 35 to 44 years old (8) 45 to 54 years old (9) 55 to 64 years old (10) 65 to 74 years old (11) 75 and older (12)
c00_sex c00_sex	A binary indicator of sex as reported on the 2000 decennial census	Male (0) Female (1)
c00_race	A categorical variable for race as reported on the 2000 decennial census	White (1) African American (2) American Indian or Alaskan Native (3) Asian (4) Other Asian or Pacific Islander (5) Some other race (6)
c00_nphu	The number of persons in housing unit as reported on the 2000 decennial census	1 to 16
hhid	Household identifier	
tract_id_str	Census tract identifier	
dim_pik	Simulated binary indicator of PIK assignment	No PIK assigned (0) PIK assigned (1)
Id	Variable that uniquely identifies individuals	

Table 1: Description of data elements

3.2. Weighting for Non-Linkage within Orleans Parish, LA

This section describes steps taken to generate weights that ensure person records with PIKs with Orleans Parish are representative of their Census tracts. First, we load the **twangRDC** package and our simulated dataset.

```
R> library(twangRDC)
R> data(nola_south)
```

Next, it is important that the variables of the dataset are coded as intended. In this case, we

convert several variables to factors so that they are properly analyzed.

```
R> nola_south$metarea = as.factor(nola_south$metarea)
R> nola_south$c00_age12 = as.factor(nola_south$c00_age12)
R> nola_south$c00_race = as.factor(nola_south$c00_race)
R> nola_south$c00_sex = as.factor(nola_south$c00_sex)
```

In a final data preparation step, we limit the dataset to Orleans Parish.

```
R> nola_only = subset(nola_south, metarea==556)
```

The goal is to generate weights to ensure that person records with PIKs are representative of their entire Census tract. That is, for each Census tract, we want to generate weights such that the person records with PIKs within the Census tract are representative of the tract's population. This is achieved by specifying linkage=TRUE, which specifies that we wish to generate nonresponse weights based on non-linkage, and strata="tract_id_str", which specifies that we wish to generate weights that are representative within strata defined by Census tracts. The formula sim_pik ~ c00_age12 + c00_race + c00_sex identifies the person records with PIK on the left hand side and the characteristics that we want to consider when estimating the weights the right hand side.

```
R> params = list(eta=.1,max_depth=5,min_child_weight=50)
R> res.pik = ps.xgb(sim_pik ~ c00_age12 + c00_race + c00_sex,
strata="tract_id_str",
data=nola_only,
params=params,
max.steps=50,
iters.per.step=100,
min.iter=1000,
id.var="id",
linkage=TRUE)
```

The parameters of the underlying xgboost model are specified in params. The id.var="id" provides a unique identifier for person records such that the generated weights can easily be merged back in with the original data. The other options specified in the ps.xgb function control how frequently the algorithm checks for convergence and how many iterations should be considered before stopping. First, iters.per.step=100 specifies that the algorithm only consider every 100-th iteration when evaluating convergence. Larger values improve computational time by reducing the number of balance evaluations, while smaller values may achieve slightly better balance. Next, min.iter=1000 specifies that at least 1000 iterations must be used before stopping for convergence. Larger values ensure that more complex models are evaluated before determining the optimal set of weights. Finally, max.steps=50 indicates that the algorithm should only evaluate balance up to 50 times before stopping. The maximum number of iterations of the xgboost model is given by max.steps*iters.per.step, which in this case is 5000. In general, this value should be large to ensure that the optimum set of weights is achieved. The default value is max.steps=Inf, which will continue adding iterations to the model until the convergence criteria is met. Due to computational concerns, we recommend testing your code with values of iters.per.step and max.steps such that the total number of iterations is small (1000 to 10000). Once you have determined the model is working as intended, set max.steps to Inf.

We now evaluate the quality of the weights that were generated from the boosted model. First, we need to ensure that a sufficient number of iterations have been used such that the

balance criteria is minimized. The plot function provides a plot of the balance criteria, which in this case is the average of the strata-specific maximum absolute standardized difference of the covariates, versus the number iterations. As shown in Figure 1, we are verifying that the algorithm has run for a sufficient number of iterations such that a clear minimum has been achieved.

```
R> plot(res.pik)
```

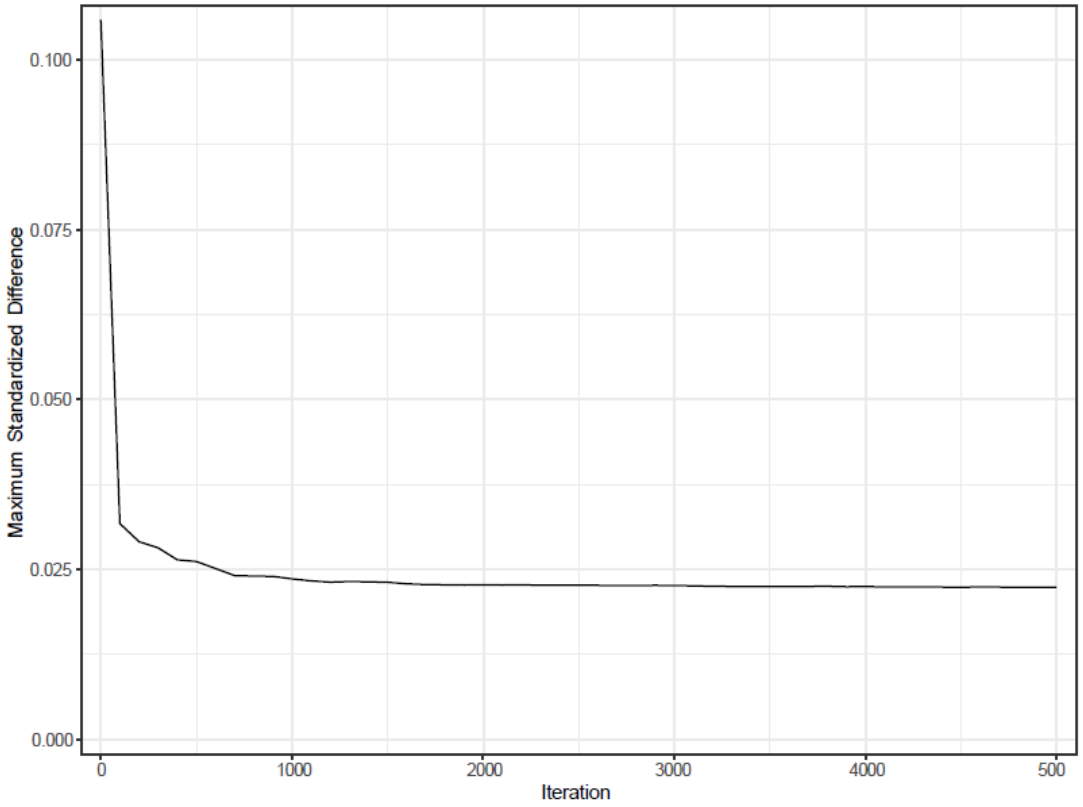


Figure 1: Convergence of gradient boosted model for non-linkage.

After evaluating convergence of the algorithm, we can access the quality of the weights using balance tables. First, the bal.table function will produce the population mean for each covariate, as well as the unweighted and the weighted mean among those with PIK. It also provides the standardized difference for each covariate before and after weighting. The goal is for the standardized differences after weighting to all be small, with many researchers recommending absolute values less than 0.1 as the target. Regardless of the specific target, standardized differences close to zero are ideal, and the quality of the balance should be assessed with the context of the specific analysis in mind. In this case, Table 2 indicates that all of our standardized differences are very close to zero. If this were not the case, we would refit the model tuning the parameters of the underlying xgboost model specified in params.

```
R> bal.table(res.pik)
```

	Population Mean	Unadjusted Mean	Adjusted Mean	Unadjusted Standardized Difference	Adjusted Standardized Difference
c00_ age12:1	0.032	0.017	0.030	0.086	0.011
c00_age12:2	0.037	0.023	0.037	0.073	-0.000
c00_age12:3	0.036	0.025	0.035	0.059	0.004
c00_age12:4	0.066	0.056	0.066	0.042	-0.000
c00_age12:5	0.060	0.065	0.060	-0.019	-0.001
c00_age12:6	0.059	0.054	0.059	0.022	-0.001
c00_age12:7	0.126	0.127	0.126	-0.004	-0.001
c00_age12:8	0.174	0.179	0.174	-0.015	-0.001
c00_age12:9	0.176	0.192	0.176	-0.042	-0.001
c00_age12:10	0.112	0.126	0.112	-0.044	-0.001
c00_age12:11	0.063	0.070	0.063	-0.028	-0.001
c00_age12:12	0.059	0.066	0.059	-0.032	-0.002
c00_race:1	0.604	0.647	0.605	-0.088	-0.003
c00_race:2	0.364	0.320	0.363	0.091	0.003
c00_race:3	0.000	0.000	0.000	-0.001	-0.000
c00_race:4	0.005	0.005	0.005	-0.009	-0.000
c00_race:5	0.012	0.013	0.012	-0.004	0.001
c00_race:6	0.015	0.015	0.015	0.001	0.001
c00_sex:0	0.512	0.511	0.511	0.003	0.003
c00_sex:1	0.488	0.489	0.489	-0.003	-0.003

Table 2: Balance of key characteristics before and after weighting for non-linkage.

Next, balance can be assessed within Census tract since the goal for this model was to generate weights such that person records with PIKs are representative of their Census tract. The type="strata" option tells bal.table to provide the maximum absolute standardized difference by the strata variable, in this case, the maximum absolute standardized difference of the covariates within each Census tract. We additionally specify include.var=TRUE, which identifies the covariate that the maximum absolute standardized difference corresponds to within each Census tract, decreasing=T, which orders the strata in decreasing order of their weighted standardized differences, and n=3, which only prints the three strata with the largest absolute standardized differences. Note that this table produces a single row for each stratum, such that the table only represents the covariate with the worst balance for each stratum (Table 3).

```
R> bal.table(res.pik,type="strata",include.var=TRUE,n=3,decreasing=T)
```

Table 3 allows us to assess which strata had the worst balance after weighting, and among those strata, which covariates were problematic. In this case, both our within strata balance and our overall population balance look excellent, as they fall below the commonly used threshold of 0.1. We conclude that the weighting has achieved our goal of making the sample representative of the population. If this were not the case, we would refit the model tuning the parameters of the underlying xgboost model specified in params.

Stratum	Unadjusted Maximum Standardized Difference	Adjusted Maximum Standardized Difference	Variable
22071012101	0.114	0.083	c00_age12:1
22071001741	0.136	0.006	c00_race:5
22071011400	0.091	0.000	c00_race:6

Table 3: Characteristic with worst balance within Census tract after weighting for non-linkage.

In a final step, we merge the nonresponse weights back onto the original dataset. The get.weights function automatically extracts the weights at the optimal iteration. The resulting object contains the weights and the ID variable specified in id.var. The weights can

then be merged back in with the original data using the id.var. Note that base R merge function is slow compared to modern alternatives. If your data is large, consider data.table or dplyr.

```
w = get.weights(res.pik) dta=merge(dta,w,by='id',all=TRUE)
```

These weights can be used in subsequent steps using standard approaches for weighted analyses or can be further calibrated using approaches such as raking or post-stratification (Särndal and Lundström 2005; Deville, Särndal, and Sautory 1993; Gelman and Carlin 2000).

4. Discussion

This article introduced the **twangRDC** R package and highlighted its use for generating non-response weights to account for non-linkage. The package contains several user-friendly functions that allow for easy estimation, diagnosis, and extraction of the weights. It provides FSRDC users access to gradient boosting models for non-linkage adjustment, which provides flexibility over logistic or probit regression models. Further, although not covered by this tutorial, the **twangRDC** package can be used to ensure that a comparison group is equivalent to a treatment group on observable characteristics. Those who want more information on the **twangRDC** R package are encouraged to see the package help files.

References

Bond B, Brown JD, Luque A, O'Hara A, *et al.* (2014). "The nature of the bias when studying only linkable person records: Evidence from the American Community Survey." *Center for Administrative Records Research and Applications Working Paper*, **8**.

Cefalu M, Sullivan J (2021). *twangRDC: Gradient Boosting for Linkage Failure in FSRDCs*. R package version 1.0, URL https://CRAN.R-project.org/package=twangRDC.

Chen T, Guestrin C (2016). "Xgboost: A scalable tree boosting system." In *Proceedings of the 22nd acm sigkdd international conference on knowledge discovery and data mining*, pp. 785–794.

Deville JC, Särndal CE, Sautory O (1993). "Generalized raking procedures in survey sampling." *Journal of the American statistical Association*, **88**(423), 1013–1020.

Gelman A, Carlin JB (2000). "Poststratification and weighting adjustments." In *In*. Citeseer.

McCaffrey DF, Ridgeway G, Morral AR (2004). "Propensity score estimation with boosted regression for evaluating causal effects in observational studies." *Psychological methods*, **9**(4), 403.

Natekin A, Knoll A (2013). "Gradient boosting machines, a tutorial." *Frontiers in neurorobotics*, **7**, 21.

Ridgeway G, McCaffrey D, Morral A, Cefalu M, Burgette L, Pane J, Griffin BA (2021). *Toolkit for Weighting and Analysis of Nonequivalent Groups: A guide to the twang package*.

Ruggles S, McCaa R, Sobek M, Cleveland L (2015). "The IPUMS collaboration: integrating and disseminating the world's population microdata." *Journal of demographic economics*, **81**(2), 203–216.

Särndal CE, Lundström S (2005). *Estimation in surveys with nonresponse.* John Wiley & Sons.

Wagner D, Lane M, *et al.* (2014). "The person identification validation system (PVS): applying the Center for Administrative Records Research and Applications'(CARRA) record linkage software." *Technical report*, Center for Economic Studies, US Census Bureau.

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